

HACKATHON PROBLEM STATEMENTS

Prompt Engineering, Generative AI and Retrieval-Augmented Generation (RAG)

S.No.	Project Title	Domain	Architecture Flow
1	AI Academic Advisor	Education	Student Query → Prompt Engineering → RAG → College Documents → Vector DB → LLM → Grounded Answer
2	CampusGPT	Education	Student Query → Query Understanding → RAG → Academic Policies → LLM → Answer + Sources
3	Personalized Learning Assistant	Education	Student Profile → Prompt → Learning Content RAG → LLM → Personalized Learning Plan
4	AI Question Paper Generator	Education	Syllabus → Prompt Engineering → Difficulty/CO Mapping → LLM → Question Paper → Validation
5	Research Paper Intelligence Assistant	Research	Research Papers → PDF Extraction → Chunking → Embeddings → Vector DB → RAG → LLM
6	CareerRAG	Education / Career	Resume → Job Description → Skill Extraction → RAG → Prompt → LLM → Skill Gap Report
7	HR Policy Assistant	HR	Employee Query → Query Rewriting → HR Documents → Vector Search → RAG → LLM → Policy Answer
8	Employee Knowledge Copilot	HR	Employee Question → Intent Detection → RAG → Company Knowledge Base → LLM → Answer
9	AI Recruitment Assistant	HR	Resume → JD → Prompt → Skill Extraction → Matching → LLM → Candidate Report
10	Employee Onboarding Assistant	HR	Employee Query → Onboarding Documents → RAG → LLM → Step-by-Step Guidance
11	Healthcare Information Assistant	Healthcare	Patient Query → Medical Documents → Embeddings → Vector DB → RAG → LLM → Evidence-Based Response
12	Patient Education Assistant	Healthcare	Patient Question → Medical Knowledge Base → Retrieval → Prompt → LLM → Simplified Explanation
13	Clinical Guidelines Assistant	Healthcare	Clinical Query → Guidelines → Hybrid Search → Reranking → RAG → LLM → Cited Response
14	Medical Research Assistant	Healthcare / Research	Research Papers → Chunking → Embeddings → Vector DB → RAG → LLM → Research Summary
15	Hospital FAQ Assistant	Healthcare	Patient Query → Hospital Documents → Semantic Search → RAG → LLM → Response
16	Loan Policy Assistant	Banking	Customer Query → Loan Documents → Vector Search → RAG → LLM → Eligibility Information
17	Banking Knowledge Copilot	Banking	Banking Query → Product Documents → Retrieval → Prompt → LLM → Grounded Answer
18	Financial Document Assistant	Finance	Financial Documents → PDF Extraction → Chunking → Embeddings → RAG → LLM → Summary
19	Insurance Policy Assistant	Insurance	Customer Query → Insurance Policies → Vector Search → RAG → LLM → Policy Explanation
20	Insurance Claim Assistant	Insurance	Claim Query → Claim Guidelines → Retrieval → Prompt → LLM → Document Checklist
21	Contract Intelligence Assistant	Legal	Contract → Document Parsing → Chunking → Vector DB → RAG → LLM → Clause Analysis
22	Legal Document Research Assistant	Legal	Legal Query → Legal Documents → Hybrid Search → Reranking → RAG → LLM
23	Legal Clause Finder	Legal	Contract → Clause Extraction → Embeddings → Similarity Search → LLM → Relevant Clauses
24	Government Services Assistant	Government Services	Citizen Query → Government Documents → RAG → LLM → Eligibility + Procedure
25	Manufacturing Knowledge Copilot	Manufacturing	Engineer Query → SOPs/Manuals → Vector Search → RAG → LLM → Technical Guidance
26	Machine Troubleshooting Assistant	Manufacturing	Machine Problem → Error Code → Manuals → Retrieval → RAG → LLM → Troubleshooting Steps
27	Intelligent SOP Assistant	Manufacturing	Operator Query → SOP Repository → Semantic Search → RAG → LLM → Procedure
28	Quality Control Knowledge Assistant	Manufacturing	Quality Issue → QC Documents → Retrieval → RAG → LLM → Recommended Procedure
29	DevRAG	IT / Software	Developer Query → Technical Docs → Embeddings → Vector DB → RAG → LLM → Code/Explanation
30	IT Troubleshooting Assistant	IT	IT Problem → Knowledge Base → Semantic Search → RAG → LLM → Troubleshooting Guide
31	API Documentation Copilot	Software Development	Developer Query → API Docs → Retrieval → Prompt → LLM → API Guidance

S.No.	Project Title	Domain	Architecture Flow
32	DevOps Knowledge Assistant	DevOps	DevOps Query → Deployment Docs → RAG → LLM → Deployment Instructions
33	Cybersecurity Policy Assistant	Cybersecurity	Security Query → Security Policies → Vector Search → RAG → LLM → Security Guidance
34	Incident Response Assistant	Cybersecurity	Incident → Security Knowledge Base → Retrieval → LLM → Response Procedure
35	Agriculture Knowledge Assistant	Agriculture	Farmer Query → Agriculture Documents → RAG → LLM → Contextual Guidance
36	Crop Knowledge Assistant	Agriculture	Crop Query → Crop Documents → Retrieval → Prompt → LLM → Crop Information
37	Travel Planning Assistant	Travel	Travel Query → Destination Knowledge Base → RAG → LLM → Personalized Itinerary
38	Hotel Operations Assistant	Hospitality	Staff Query → Hotel SOPs → Vector Search → RAG → LLM → Operational Guidance
39	Supply Chain Knowledge Assistant	Supply Chain	Supply Chain Query → SOPs → Retrieval → RAG → LLM → Recommended Procedure
40	Construction Project Assistant	Construction	Project Query → Specifications/SOPs → RAG → LLM → Project Guidance
41	Energy Operations Copilot	Energy	Engineer Query → Equipment Manuals → Retrieval → RAG → LLM → Operational Guidance
42	Automotive Service Assistant	Automotive	Vehicle Issue → Service Manuals → Error Codes → RAG → LLM → Service Procedure
43	E-Commerce Product Assistant	E-Commerce	Customer Need → Product Catalog → Semantic Search → RAG → LLM → Product Comparison
44	Customer Support RAG Assistant	Customer Service	Customer Query → FAQ/Knowledge Base → Retrieval → RAG → LLM → Support Response
45	Enterprise Knowledge Copilot	Enterprise	Employee Query → Enterprise Documents → Hybrid Search → Reranking → RAG → LLM
46	Multi-Agent RAG Assistant	Multi-Domain	User Query → Supervisor Agent → Specialist Agents → RAG → Tools → LLM → Final Response
47	Knowledge Graph RAG Assistant	Multi-Domain	Documents → Entity Extraction → Knowledge Graph + Vector DB → Hybrid Retrieval → LLM
48	Hybrid RAG Intelligence System	Multi-Domain	Query → Keyword Search + Vector Search → Reranker → Context → LLM → Grounded Answer
49	Hallucination-Aware RAG Assistant	Multi-Domain	Query → RAG → LLM → Fact Verification → Citation Check → Corrected Answer
50	Prompt Optimization Assistant	AI / GenAI	Task → Multiple Prompts → LLM Evaluation → Prompt Selection → Optimized Prompt
51	AI Document Intelligence Platform	Multi-Domain	Documents → OCR/Parsing → Chunking → Embeddings → RAG → LLM → Structured Output

Suggested project progression: Prompt Engineering → Generative AI → Basic RAG → Advanced RAG → Agentic RAG → Knowledge Graph RAG.